

ECE 4375, Final Exam Review Material

1. Consider the differential equations,

$$\ddot{y}(t) - \dot{y}(t) + 6y(t) = u(t)$$

$$\ddot{y}(t) + 5\dot{y}(t) + 2y(t) - \dot{u}(t) + u(t) = 0$$

Find the transfer functions, $\frac{Y(s)}{U(s)}$. What assumptions must be made regarding the initial conditions? Is the system stable? What are the initial and final values of the system when the input is a unit step?

Now, take the $\frac{Y(s)}{U(s)}$ transfer function and apply unity negative feedback around it. How has the feedback changed the stability? What are the new initial and final values when the input is a unit step?

Finally, transform the $\frac{Y(s)}{R(s)}$ transfer function back into a differential equation.

2. Given the block diagram in Figure 1, perform the following tasks

1. Determine the BIBO stability with the Routh Array
2. Sketch the Root Locus when G_c is a P, PI, PD, and PID

When appropriate find:

- (a) σ_i
 - (b) Θ_k
 - (c) BA and BI
 - (d) gain K at the BA and BI points
 - (e) critical gain for stability, K_{cr}
3. ~~Determine the frequency response parameters (gain margin, phase margin, and bandwidth) when $G_c = 1$~~
4. Determine the steady-state error for position, velocity, and acceleration

When,

A.

$$G(s) = \frac{s^2 + 2s + 2}{(s^2 - 11s + 24)(s^2 + 7s)}$$

B.

$$G(s) = \frac{s - 9}{(s - 3)(s^2 - 6s + 5)(s^2 + 2s + 5)}$$

C.

$$G(s) = \frac{1}{(s + 1)(s + 10)}$$

D.

$$G(s) = \frac{s + 0.1}{(s + 1)(s + 10)}$$

E.

$$G(s) = \frac{s+1}{s^2(s+10)^2}$$

F.

$$G(s) = \frac{900}{s^2 + 15s + 900}$$

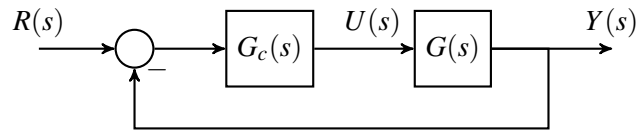


Figure 1: Generic block diagram.

3. Given the 2nd order model for a DC-servo motor, $\frac{\omega_m(s)}{V_a(s)}$ where:

1. $R_a = 1.1$
2. $L_a = 0.0001$
3. $K_m = 12$
4. $B = 0.75$
5. $J_T = 0.85$
6. $K_b = 0.125$

Find the transfer function, $\frac{I_a(s)}{V_a(s)}$. What is the steady-state current given a step input?

unit = 1/s
impulse = 1

FVT: $y(\infty) = \lim_{s \rightarrow 0} sY(s)$, BIBO stable

IVT: $y(0) = \lim_{s \rightarrow \infty} sY(s)$

$\frac{1}{s} = \infty$ $\frac{1}{s} = 0$

Quadratic: $-b \pm \sqrt{b^2 - 4ac}$
divide by s^2 from denom $2a$

R → F

time → Laplace	$\delta(t)$	1	$\sin(bt)$	$b/(s^2+b^2)$	$x(at)$	$\frac{1}{a}X(\frac{s}{a})$
	1	$1/s$	$\cos(bt)$	$s/(s^2+b^2)$	$x(t-T)$	$e^{-Ts}X(s)$
	e^{-at}	$1/[s+a]$	$e^{-at}\sin(bt)$	$b/[(s+a)^2+b^2]$	x'	$sX(s)-x(0^-)$
	t^n	$n!/s^{n+1}$	$e^{-at}\cos(bt)$	$(s+a)/[(s+a)^2+b^2]$	x''	$s^2X(s)-sx(0^-)-x'(0^-)$
	te^{-at}	$1/(s+a)^2$			$\int_0^t x(t')dt$	$\frac{1}{s}X(s)$

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$$s^2 Y(s) - s \cdot Y(0) - Y'(0) - s Y(s) - Y(0) + 6 Y(s) = U(s)$$

$$Y(s)[s^2 - s + 6] = U(s) \rightarrow \frac{Y(s)}{U(s)} = \frac{1}{s^2 - s + 6}$$

$$s^2 - s + 6 = 0 \rightarrow s = 0.5 + 2.397j, 0.5 - 2.397j$$

ICs must be zero.

unstable poles $\therefore$ system is unstable

$$\underline{\text{FV:}} \quad y(\infty) = \lim_{s \rightarrow 0} s Y(s) = \frac{1}{s} \cdot \frac{s}{s^2 - s + 6} = \frac{1}{s^2 - s + 6} = \frac{1}{6}$$

$$\underline{\text{IV:}} \quad y(0) = \lim_{s \rightarrow \infty} s Y(s) = \frac{1}{\infty} = 0$$

$$s^2 Y(s) + 5s Y(s) + 2Y(s) = s U(s) - U(s)$$

$$Y(s)[s^2 + 5s + 2] = U(s)[s - 1]$$

$$\frac{Y(s)}{U(s)} = \frac{s - 1}{s^2 + 5s + 2}$$

$$s^2 + 5s + 2 = 0$$

$$s = -4.56, -0.438 \quad \text{system is stable}$$

$$\underline{\text{FV:}} \quad y(\infty) = \lim_{s \rightarrow 0} s Y(s) = \frac{(s - 1)}{s^2 + 5s + 2} = \frac{-1}{2} = -\frac{1}{2}$$

$$\underline{\text{IV:}} \quad y(0) = \lim_{s \rightarrow \infty} s Y(s) = \frac{s - 1}{s^2 + 5s + 2} = \frac{1 - \frac{1}{s}}{s^2 - s + 6} = \frac{1 - 0}{\infty^2 - 0 + 6} = 0$$

$$1a \quad \frac{Y(s)}{R(s)} = \frac{\frac{1}{s^2 - s + 6}}{1 + \frac{1}{s^2 - s + 6}} = \frac{1}{s^2 - s + 7} = 0.5 + 2.598j, 0.5 - 2.598j$$

This is still unstable as the real part of the poles are still in the OLHP

$$y''(t) - y'(t) + 7y(t) = r(t)$$

FV is now $\frac{1}{7}$, and IV is still 0.

$$1b \quad \frac{Y(s)}{R(s)} = \frac{s-1}{s^2 + 5s + 2} = \frac{s-1}{s^2 + 6s + 1}$$

$$s^2 + 6s + 1 = 0$$

$$s = -5.826, -0.171$$

still stable

FV is now -1 and IV is 0

$$y''(t) + 6y'(t) + y(t) = r'(t) - r(t)$$

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6. $K_b = 0.125$

$$\frac{\omega_m}{V_a} = \frac{K_m}{L_a J_T s^2 + L_a B s + R_a J_T s + R_a B + K_m K_b}$$

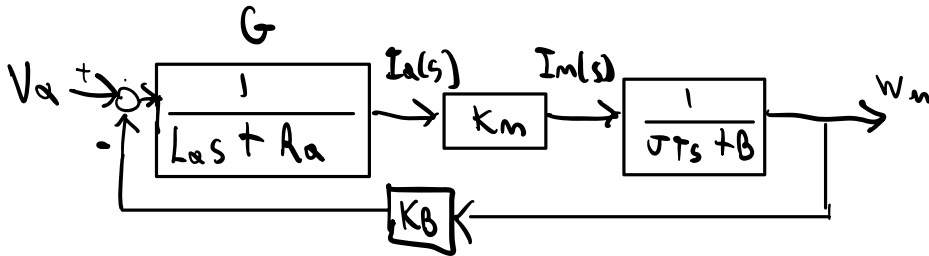
$$\cdot X = V_a(s) - \omega_m \cdot K_b$$

$$\cdot \omega_m = \frac{X \cdot G \cdot K_b}{J_T s + B}$$

$$\cdot I_a = X \cdot G$$

$$\cdot \omega_m = \frac{I_a K_m}{J_T s + B}$$

Find the transfer function, $\frac{I_a(s)}{V_a(s)}$. What is the steady-state current given a step input?



$$\frac{I_a}{G} = V_a(s) - \frac{I_a K_b K_m}{J_T s + B}$$

$$I_a(s) \left[\frac{1}{G} + \frac{K_b}{J_T s + B} \right] = V_a(s) \rightarrow \frac{I_a(s)}{V_a(s)} = \frac{1}{\frac{1}{G} + \frac{K_b K_m}{J_T s + B}}$$

$$\frac{I_a(s)}{V_a(s)} = \frac{J_T s + B}{s^2 (L_a R_a J_T) + s (L_a R_a B) + K_b K_m}$$

$$\frac{I_a(s)}{V_a(s)} = \frac{10,000 s + 8023.53}{s^2 + 11,000.9 s + 27,352.9} \quad \left(\rightarrow V_a(s) = \frac{1}{s} \right)$$

$$\text{FVT: } \lim_{s \rightarrow 0} s I_a(s) = 0.322581$$

2. Given the block diagram in Figure 1, perform the following tasks

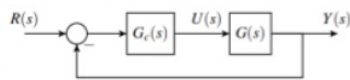


Figure 1: Generic block diagram.

A.

$$G(s) = \frac{s^2 + 2s + 2}{(s^2 - 11s + 24)(s^2 + 7s)}$$

$$P = K_P$$

$$PI = K_P + \frac{K_I}{s}$$

$$PD = K_P + K_D s$$

$$PID = K_P + \frac{K_I}{s} + K_D s$$

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1

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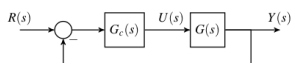


Figure 1: Generic block diagram.

$$A. \perp G(s) = \frac{s^2 + 2s + 2}{s^4 - 4s^3 - 53s^2 + 168s}$$

RA:

$$\begin{array}{c|ccc} s^4 & 1 & -53 & 0 \\ s^3 & -4 & 168 & \\ s^2 & X_2 & & \\ s^1 & X_1 & & \\ s^0 & X_5 & & \end{array}$$

sign change so system is unstable

P Controller: $G_c = K$, $H = 1$

$$L = G_c \cdot G \cdot H = \frac{s^2 + 2s + 2}{s^4 - 4s^3 - 53s^2 + 168s}$$

$$\sigma_i = \frac{[-7 + 0 + 3 + 8] - [-1 + j - 1 - j]}{121}$$

$$\sigma_i = \frac{6}{2} = 3$$

$$\sigma_0 = \frac{(2(0) + 1) \pi}{2} = \frac{\pi}{2}$$

$$\underline{BA/BI} \Rightarrow N_L^T O_L - O_L^T N_L = 0$$

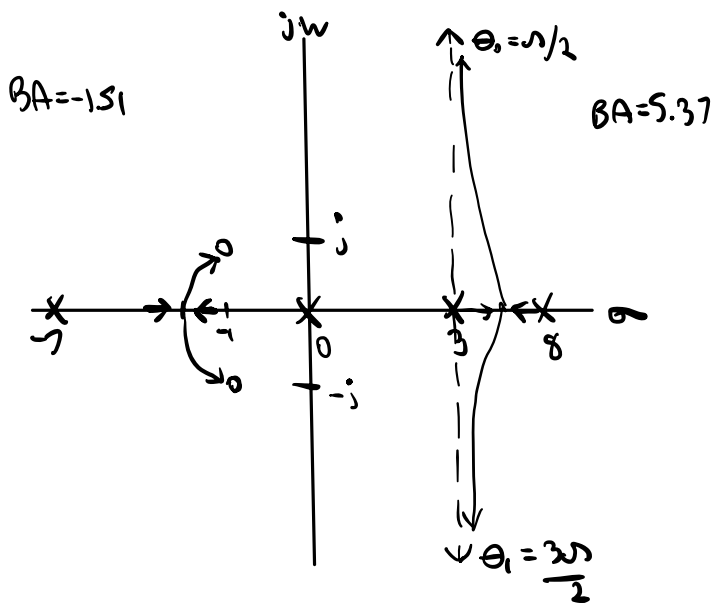
$$\sigma_1 = \frac{(2(1) + 1) \pi}{2} = \frac{3\pi}{2}$$

$$(2s + 2)(s^4 - 4s^3 - 53s^2 + 168s) - (4s^3 - 12s^2 - 106s + 168)(s^2 + 2s + 2)$$

$$s = -1.508, 0.76, 5.374$$

$$K_{BA} = \left. -\frac{O_L}{N_L} \right| = \frac{-(s^4 - 4s^3 - 53s^2 + 168s)}{s^2 + 2s + 2} \quad \begin{array}{l} 262.16 \\ 9.96 \end{array}$$

$s = -1.51, 5.37$



PI:

• K_I : add one zero // $K_I = \frac{K_I}{K_p}$ • $K_{cr} = K_p$ (do RA for K_p)

PD:

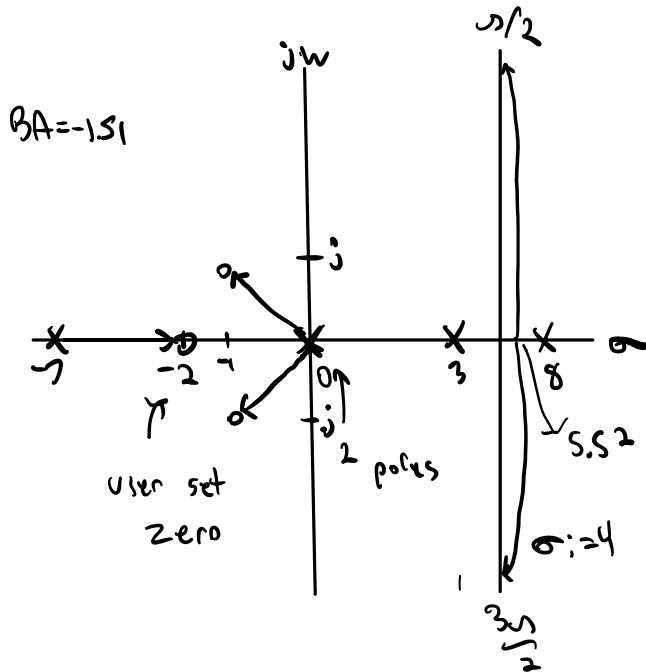
• K_p : add one zero // $K_p = \frac{K_p}{K_0}$ • $K_{cr} = K_0$

PID:

• K_I, K_p : two real zeros on a complex conjugate pair // $K_I = \frac{K_I}{K_p}, K_p = \frac{K_p}{K_0}$ • $K_{cr} = K_0$

PI: $G_c = K_p + \frac{K_I}{s} = \frac{K_p s + K_I}{s}$

$K_{cr} = K_p$



$K_I^* = -2$

$$\sigma_1 = \frac{[-7 + 0 + 3 + 8] - [-1 + j - 1 - j + -2]}{121} = 4$$

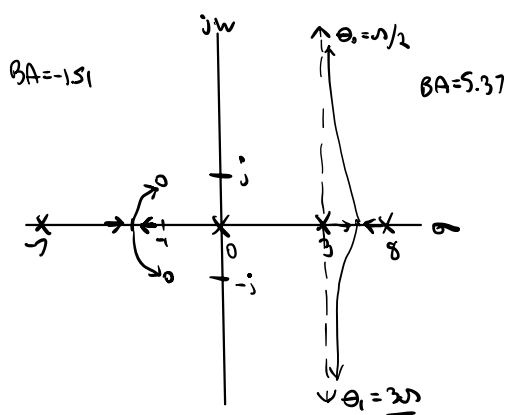
$L = G_c \cdot G \cdot H =$

$$\frac{(s^2 + 2s + 2)}{s^4 - 4s^3 - 53s^2 + 168s} \cdot \frac{K_p(s+2)}{s} =$$

$$L = \frac{(s^2 + 2s + 2)(s+2)K_p}{(s^4 - 4s^3 - 53s^2 + 168s)s}$$

$N_L^* O_L - O_L^* N_L = 0$

BA at S.S.2

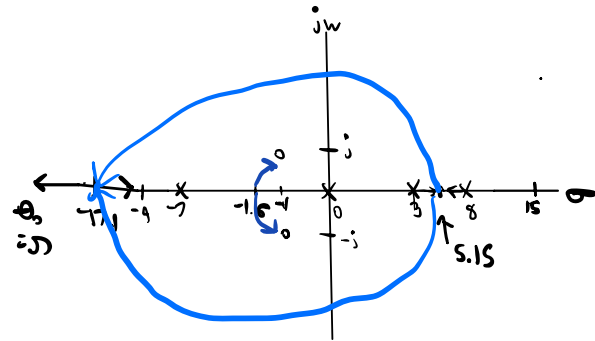


PD Controller

$$G_c = K_p + K_D \cdot s = K_D(s + K_p^{\wedge}) \quad K_{cr} = K_D$$

$$G_c = K_D(s + K_p^{\wedge}) \quad K = 0, 1,$$

$K_p^{\wedge} = 9$ so that
the locus at the
BA converges to it



$$\sigma_1 = \frac{[-7 + 0 + 3 + 8] - [-1 + j - 1 - j - 9]}{14 - 3} = 15$$

$$\sigma_0 = \frac{(2(0) + 1)j}{14 - 3} = j$$

$$L = G_c \cdot G \cdot H = \frac{K_D(s + 9)(s^2 + 2s + 2)}{(s^4 - 4s^3 - 53s^2 + 166s)}$$

$$N_L^T D_L - D_L^T N_L = 0$$

$$s = -17.12, -1.61244, 0.7047, 5.15977$$

$$\frac{K_{BA}}{BI} \quad s = -17.12 \rightarrow \left. \frac{-D_L}{N_L} \right|_{s = BI @ -17.12} = -1.61$$

$$\left. \frac{-D_L}{N_L} \right|_{s = BA = 5.15} = 0.697$$

$$A(s) = D_L + K N_L = s^4 - 4s^3 - 53s^2 + 166s + K[(s + 9)(s^2 + 2s + 2)]$$

s^4	1	$11K - 53$	$18K$
s^3	$K - 4$	$20K + 168$	
s^2	X_{31}		
s^1	X_{41}		
s^0	X_{51}		

$$X_{31} = \frac{(K - 4)(11K - 53) - (20K + 168)}{K - 4}, K > 4$$

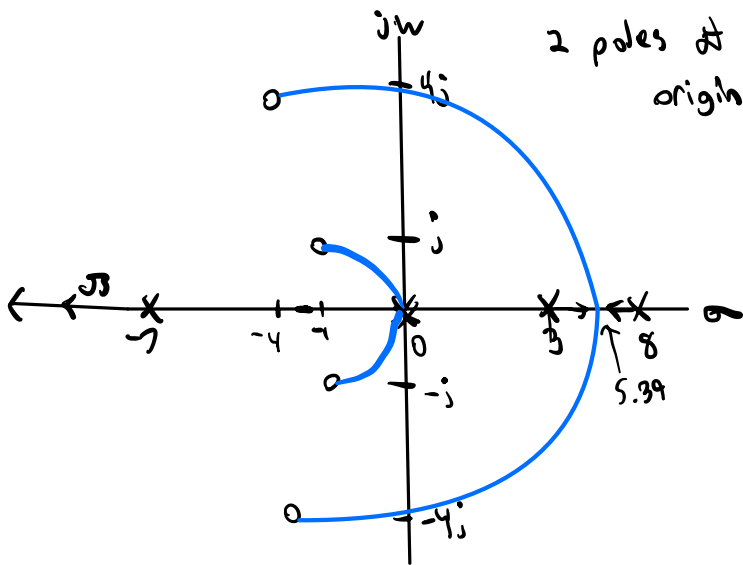
$$X_{41} = \frac{X_{31}(20K + 168) - 18(K - 4)}{X_{31}}, K > 0.24$$

$$K_{cr} = K_D > 0.43$$

$$K_{cr} = K_D = 0.4$$

$$K_p = K_p^{\wedge} \cdot K_0 = 9 \cdot 10.4 = 93.6$$

PID Controller: $G_c = K_p + \frac{K_I}{s} + K_D s = \frac{K_0(s^2 + K_p^{\wedge}s + K_I^{\wedge})}{s}$ $K_p^{\wedge} = \frac{K_p}{K_0}$



PID:
 $\cdot K_I^{\wedge}, K_p^{\wedge}$: two real zeros on a complex conjugate pair // $K_I^{\wedge} = \frac{K_I}{K_0}$, $K_p^{\wedge} = \frac{K_p}{K_0}$
 $\cdot$ pole at origin

$$(s+4+4j)(s+4-4j) = s^2 + K_p^{\wedge}s + K_I^{\wedge}$$

$$K_p^{\wedge} = 8$$

$$K_I^{\wedge} = 32$$

$$\sigma_i = \frac{(-7 + -3 + 0 + 8) - (-1 + -1 -4 -4)}{15 - 4}$$

$$\sigma_i = 14$$

$$\sigma_o = \frac{(210 + 1)j}{|n-m|} \quad K=0$$

$$L = G_c \cdot G \cdot H = \frac{K_0(s^2 + 8s + 32)}{s} \cdot \frac{(s^2 + 2s + 2)}{s^4 - 4s^3 - 53s^2 + 168s}$$

$$N_L' D_L - N_L D_L' = 0 \quad s = [-2/15, 0, 1/10, 5.39]$$

BA

$$K_{BA} = - \frac{D_L}{N_L} \bigg|_{s=5.39} \rightarrow 0.515$$

$$s^4(K-4) \quad s^3(6K-53)$$

$$A(s) = D_L + K N_L = s^5 - 4s^4 - 53s^3 + 168s^2 + Ks^4 + 10Ks^3 + 50Ks^2 + 60Ks + 64K$$

s^5	1	$10K-53$	$60K$
s^4	$K-4$	$50K+168$	$64K$
s^3	X_{31}	X_{32}	0
s^2	X_{41}	X_{42}	0
s^1	X_{51}	0	0
s^0	X_{61}	0	0

$$X_{31} = \frac{(K-4)(60K-53) - (50K+168)}{K-4} \quad X_{31} > 13.985$$

$$X_{32} = \frac{(K-4)(60K) - 64K}{K-4} > 0 \quad K > 4.8$$

$$\underline{K_0 > 14.9}$$

$$X_{41} = \dots > 0$$

$$K_I = \hat{K}_I \cdot K_0 = 476.8$$

$$K_P = \hat{K}_P \cdot K_0 = 119.2$$